MAE 5083 EQUATION SHEET

Fundamental Properties

Complex Numbers: $\underline{A} = x + jy = |\underline{A}|e^{j\theta}$

$$|\underline{A}| = \sqrt{x^2 + y^2}; \quad \theta = \tan^{-1}(y/x)$$

$$e^{\pm j\omega t} = \cos(\omega t) \pm j\sin(\omega t)$$

Wave Properties:

$$\omega = 2\pi f \qquad f = 1/T$$

$$c = \lambda/T \qquad |\vec{\underline{k}}| = 2\pi/\lambda$$

$$T = 2\pi/\omega \qquad s \equiv (\rho - \rho_o)/\rho_o$$

$$\underline{z} = \underline{p}/\vec{\underline{u}} \cdot \hat{n}_{in}$$

Taylor Expansion:

$$f(x) = \sum_{n=0}^{\infty} \left(\frac{\partial^n f}{\partial x^n}\right)\Bigg|_{x=a} \frac{(x-a)^n}{n!}$$

Driven-Damped Oscillations

$$m\frac{d^2x}{dt^2} + R\frac{dx}{dt} + Sx = f(t) = Re\left\{\underline{f}\right\}$$

If $\underline{f} = Fe^{j\omega t}$ and $\underline{x} = \underline{A}e^{j\omega t}$:

$$\underline{x}(t) = \frac{Fe^{j\omega t}}{(S - \omega^2 m) + j(\omega R)}$$

$$\underline{u}(t) = \frac{Fe^{j\omega t}}{R + j(\omega m - S/\omega)}$$

$$\underline{Z}_m = \frac{\underline{f}}{\underline{u}} = R + j(\omega m - S/\omega)$$

$$\Pi = \frac{1}{T}\int_0^T fu\,dt$$

Resonance: $\omega_o = \sqrt{S/m}$

$$Q = \frac{\omega_o}{\omega_2 - \omega_1}$$

Vibrating Spring with Fixed Ends

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 y}{\partial t^2} \quad \text{with} \quad c = \sqrt{T/\rho_L}$$

$$k_n = \omega_n/c = 2\pi f_n/c$$

$$y(x,t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right)\cos\left(\frac{n\pi ct}{L}\right)$$

$$A_n = \frac{2}{L}\int_0^L y(x,0)\sin\left(\frac{n\pi x}{L}\right)dx$$

Acoustic Wave Equation

Thermal Equation of State: $\mathcal{P} = \rho r T_K$

$r = R_u/M_w \qquad R_u = 8.3144$ J/mole·K

Isothermal Equation of State:

$$\mathcal{P}/\mathcal{P}_o = \rho/\rho_o \quad \text{or} \quad p/\mathcal{P}_o = s$$

Adiabatic Equation of State:

$$\frac{\mathcal{P}}{\mathcal{P}_o} = \left(\frac{\rho}{\rho_o}\right)^\gamma$$

$p = \mathcal{B}s$ (1st order approx for $s \ll 1$)

$$\mathcal{B} = \rho_o\left(\frac{\partial\mathcal{P}}{\partial\rho}\right)_{adiabtic}$$

Conservation of Mass (continuity)

$$\frac{\partial\rho}{\partial t} + \vec{\nabla}\cdot(\rho\vec{u}) = 0$$

Conservation of Momentum (Euler)

$$\frac{\partial\vec{u}}{\partial t} + (\vec{u}\cdot\vec{\nabla})\vec{u} + \frac{1}{\rho}\vec{\nabla}\mathcal{P} = 0$$

Linearized Governing Equations ($s \ll 1$):

$$p = \rho_o c^2 s$$

$$\frac{1}{\rho_o c^2}\frac{\partial p}{\partial t} + \vec{\nabla}\cdot\vec{u}' = 0$$

$$\rho_o\frac{\partial\vec{u}'}{\partial t} + \nabla p = 0$$

Lossless Acoustic Wave Equation $(\vec{U} = 0)$

$$\nabla^2 p = \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} \quad \text{where} \quad c = \sqrt{\mathcal{B}/\rho_o}$$

$$c = \sqrt{\mathcal{B}/\rho_o} = \sqrt{\left(\frac{\partial\mathcal{P}}{\partial\rho}\right)_{adiabtic}} = \sqrt{\frac{\gamma\mathcal{P}_o}{\rho_o}}$$

Energy Density & Intensity

$$e_T = \frac{1}{2}|\vec{u}|^2 + \frac{p^2}{2\rho_o^2 c^2}$$

$$\mathcal{E}_i = \frac{1}{2}\rho_o\left(|\vec{u}|^2 + \frac{p^2}{\rho_o^2 c^2}\right) \qquad \mathcal{E} = \frac{1}{T}\int_0^T \mathcal{E}_i dt$$

$$\vec{I}_i = p\vec{u} \qquad \vec{I} = \frac{1}{T}\int_0^T \vec{I}_i\,dt$$

$$p_e = \sqrt{\frac{1}{T}\int_0^T p^2\,dt} \qquad (p_e)_{harmonic} = \frac{|\underline{p}|}{\sqrt{2}}$$

Decibel Scales

$$IL = 10\log_{10}\left(\frac{|\vec{I}|}{|\vec{I}_{ref}|}\right)$$

$$\left(\vec{I}_{ref}\right)_{air} = 10^{-12}\ \mathrm{W/m^2}$$

$$SPL = 20\log_{10}\left(\frac{p_e}{\mathcal{P}_{ref}}\right)$$

$$\left(\mathcal{P}_{ref}\right)_{air} = 20\ \mu\mathrm{Pa}$$

Harmonic Plane Waves

$$\underline{p} = \underline{A}e^{j(\omega t - \vec{k}\cdot\vec{r})} = |\underline{A}|e^{j(\omega t - \vec{k}\cdot\vec{r}+\theta)}$$

$$\vec{k} = (k_x, k_y, k_z) \qquad\qquad \vec{r} = (x, y, z)$$

$$|\vec{\underline{k}}| = \omega/c$$

$$\omega t - k_x x_{crest} - k_y y_{crest} - k_z z_{crest} + \theta = 2\pi m$$

$$\frac{d\vec{r}_{crest}}{dt} = \frac{\vec{k}}{|\vec{\underline{k}}|}c$$

$$\vec{\underline{u}} = \frac{\vec{k}}{|\vec{\underline{k}}|}\frac{\underline{p}}{\rho_o c} \qquad\qquad \underline{z}_{plane} = \rho_o c$$

1D Plane Waves; $\vec{k} = (k_x, 0{,}0)$

$$\underline{p} = \underline{A}e^{j(\omega t - k_x x)}, \qquad\qquad \underline{A} = |\underline{A}|e^{j\theta}$$

$$\vec{\underline{u}} = \hat{x}\frac{\underline{p}}{\rho_o c}$$

$$\mathcal{E}_{plane} = \frac{p_e^2}{\rho_o c^2}, \qquad\qquad \vec{I}_{plane} = \hat{x}\frac{p_e^2}{\rho_o c}$$

1D Uniform Moving Medium: $\vec{U} = U_x\hat{x} = \text{constant}$

$$\left(\frac{\partial}{\partial t} + \vec{U}\cdot\nabla\right)^2 p - c^2\nabla^2 p = 0$$

$$k_\pm = \frac{\omega}{U_x \pm c}$$

1D Harmonic Spherical Waves

$$\frac{\partial^2(rp)}{\partial r^2} = \frac{1}{c^2}\frac{\partial^2(rp)}{\partial t^2} \qquad \text{for} \qquad r \neq 0$$

$$\underline{p} = \frac{\underline{A}}{r}e^{j(\omega t - kr)}$$

$$\vec{\underline{u}} = \frac{\hat{r}}{\rho_o c}\left(1 + \frac{1}{jkr}\right)\underline{p}$$

$$\underline{z}_{sphere} = \rho_o c\left(\frac{k^2r^2}{k^2r^2+1} + j\frac{kr}{k^2r^2+1}\right)$$

Transmission Phenomena

$$\underline{R} = \mathbb{P}_r/\mathbb{P}_i \qquad\qquad \underline{T} = \mathbb{P}_t/\mathbb{P}_i$$

$$R_I = |\vec{I}_r|/|\vec{I}_i| \qquad\qquad T_I = |\vec{I}_t|/|\vec{I}_i|$$

Plane Waves, Normal Incident, 2-fluid

$$\underline{p}_1 = \underline{\mathbb{P}}_i e^{j(\omega t - k_1 x)} + \underline{\mathbb{P}}_r e^{j(\omega t - k_1 x)}$$

$$\underline{p}_2 = \underline{\mathbb{P}}_t e^{j(\omega t - k_2 x)}$$

$$\underline{R} = \frac{\rho_2 c_2 - \rho_1 c_1}{\rho_2 c_2 + \rho_1 c_1} \qquad\qquad \underline{T} = \frac{2\rho_2 c_2}{\rho_1 c_1 + \rho_2 c_2}$$

$$R_I = |\underline{R}|^2 \qquad\qquad T_I = (\rho_1 c_1/\rho_2 c_2)|\underline{T}|^2$$

Plane Waves, Normal Incident, 3-fluid

$$\underline{p}_1 = \underline{\mathbb{P}}_i e^{j(\omega t - k_1 x)} + \underline{\mathbb{P}}_r e^{j(\omega t - k_1 x)}$$

$$\underline{p}_2 = \underline{A}e^{j(\omega t - k_2 x)} + \underline{B}e^{j(\omega t - k_2 x)}$$

$$\underline{p}_3 = \underline{\mathbb{P}}_t e^{j(\omega t - k_3 x)}$$

$$\underline{R} = \frac{\left(1 - \frac{\rho_1 c_1}{\rho_3 c_3}\right)\cos(k_2 L) + j\left(\frac{\rho_2 c_2}{\rho_3 c_3} - \frac{\rho_1 c_1}{\rho_2 c_2}\right)\sin(k_2 L)}{\left(1 + \frac{\rho_1 c_1}{\rho_3 c_3}\right)\cos(k_2 L) + j\left(\frac{\rho_2 c_2}{\rho_3 c_3} + \frac{\rho_1 c_1}{\rho_2 c_2}\right)\sin(k_2 L)}$$

$$R_I = |\underline{R}|^2 \qquad\qquad T_I = 1 - |\underline{R}|^2$$

$$r_1 = \rho_1 c_1, \qquad r_2 = \rho_2 c_2, \qquad r_3 = \rho_3 c_3$$

$$T_I = \frac{4}{2 + \left(\frac{r_3}{r_1} + \frac{r_1}{r_3}\right)\cos^2 k_2 L + \left(\frac{r_2^2}{r_1 r_3} + \frac{r_1 r_3}{r_2^2}\right)\sin^2 k_2 L}$$

Fluid Properties

Air

$$\gamma_{air} = 1.40$$

$$r_{air} = 287.06\ \mathrm{J/kg\cdot K}$$

$$c_{ideal\ gas} = \sqrt{\gamma r T_K}$$

Notation

c	speed of sound
p	acoustic pressure = $p = \mathcal{P} - \mathcal{P}_o$
$\mathcal{P}$	thermodynamic pressure
$\mathcal{P}_o$	equilibrium thermodynamic pressure
$\vec{r}$	position vector = (x, y, z)
s	condensation $\equiv (\rho - \rho_o)/\rho_o$
$\vec{u}$	fluid (medium) particle velocity
$\vec{u}'$	velocity fluctuation = $\vec{u} - \vec{U}$
$\vec{U}$	mean fluid particle velocity
ρ	fluid (medium) density
ρ_o	equilibrium fluid density
$\rho_o c$	characteristic impedance
$\vec{\xi}$	particle displacement from equilibrium